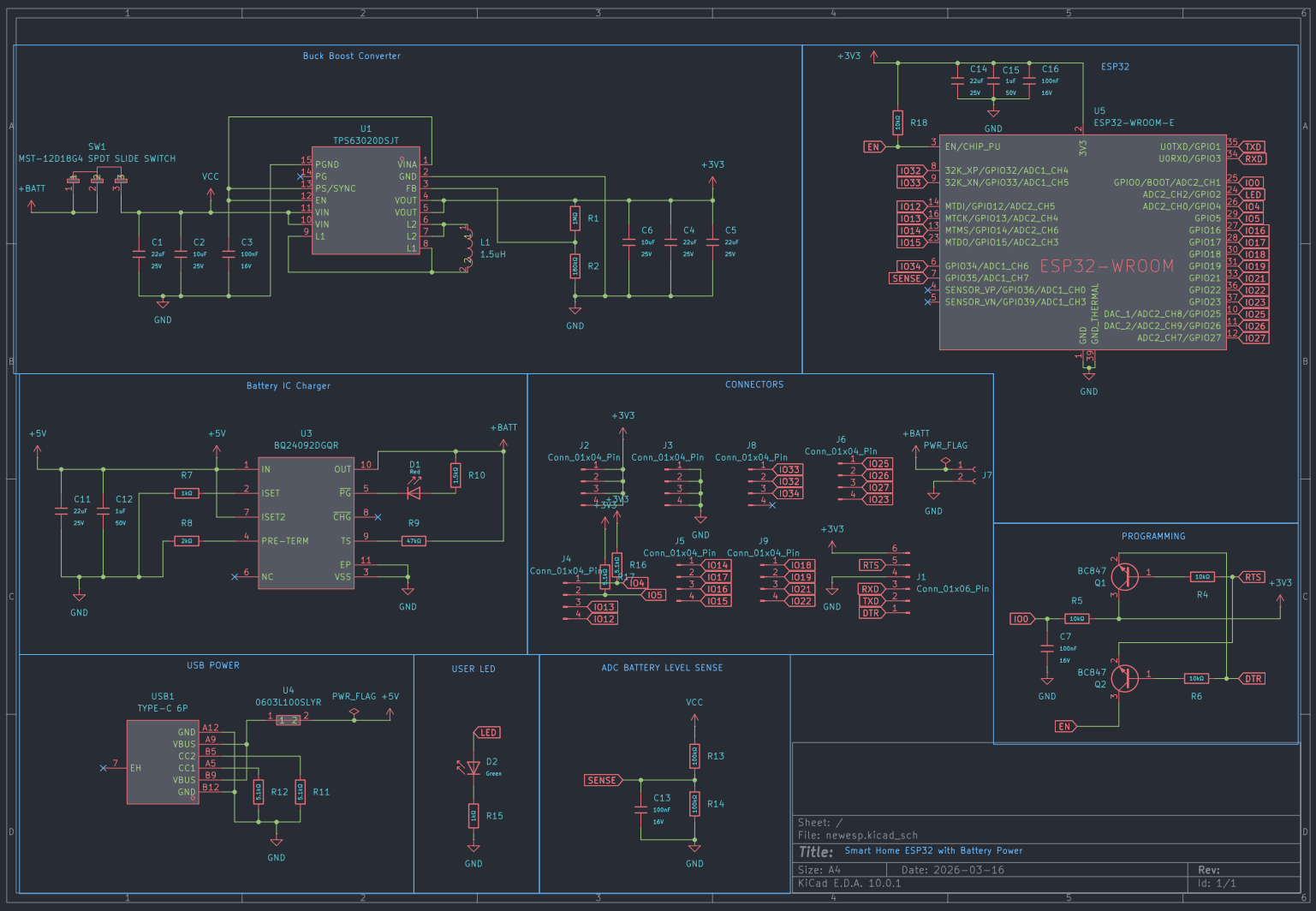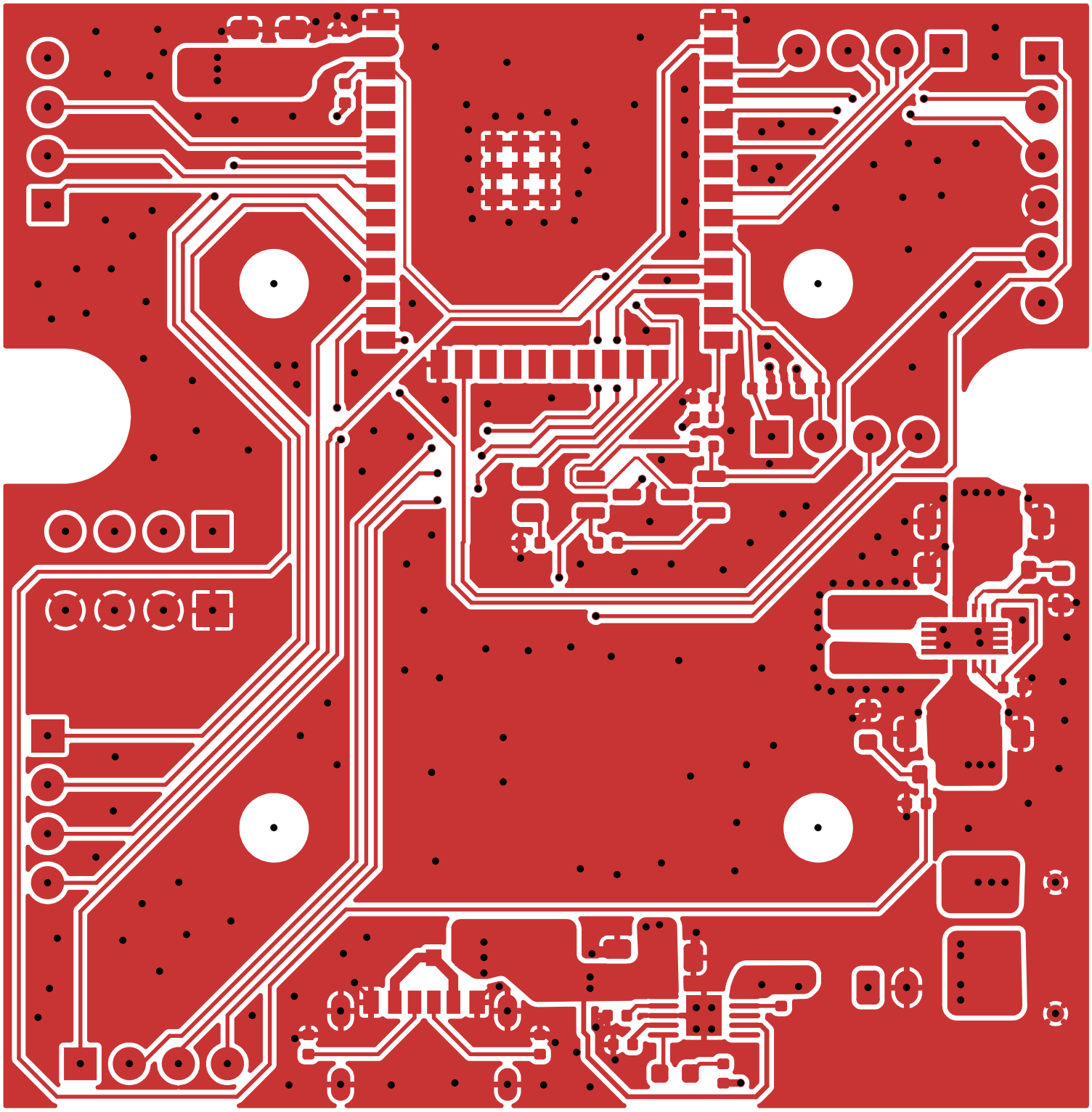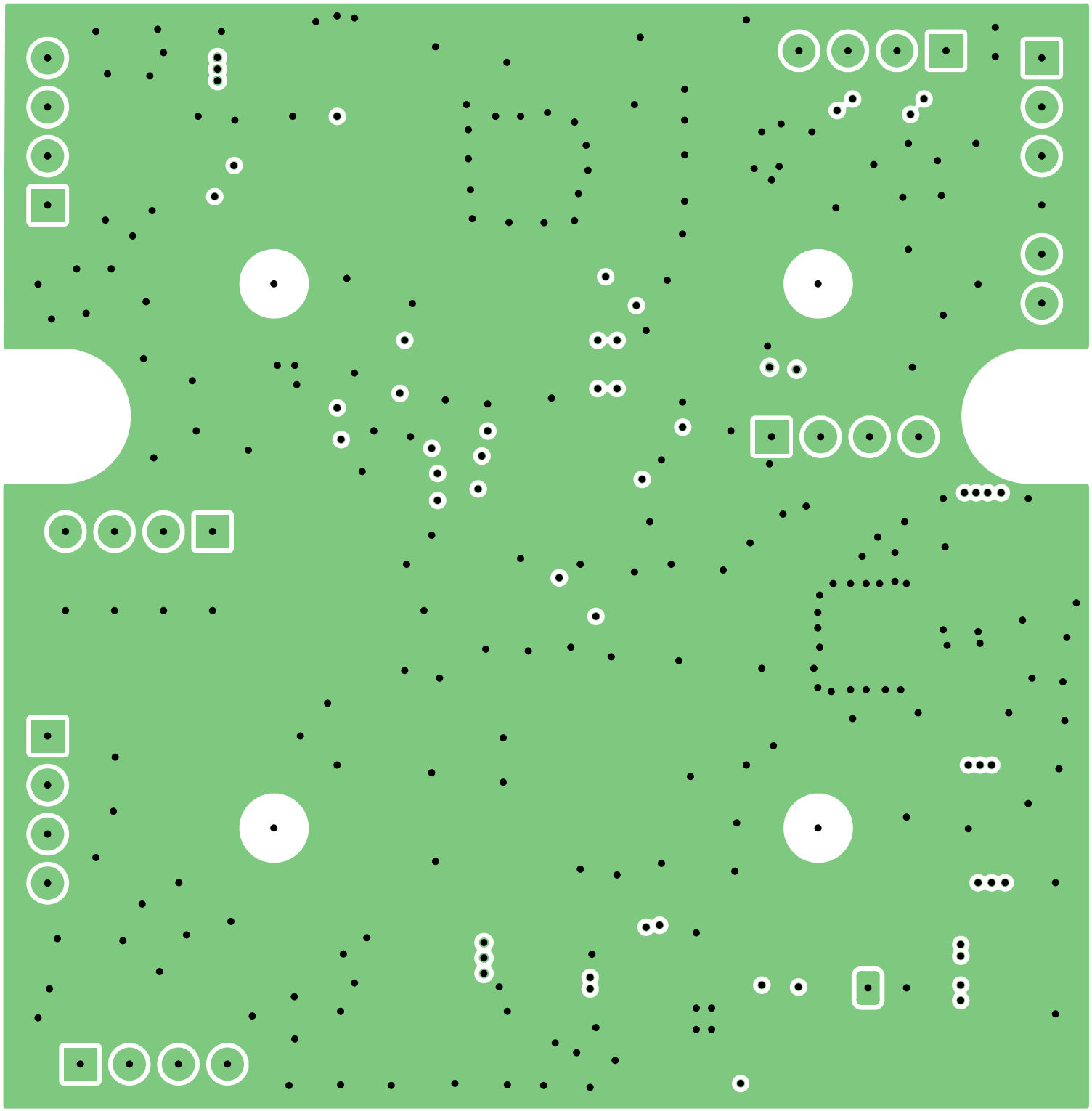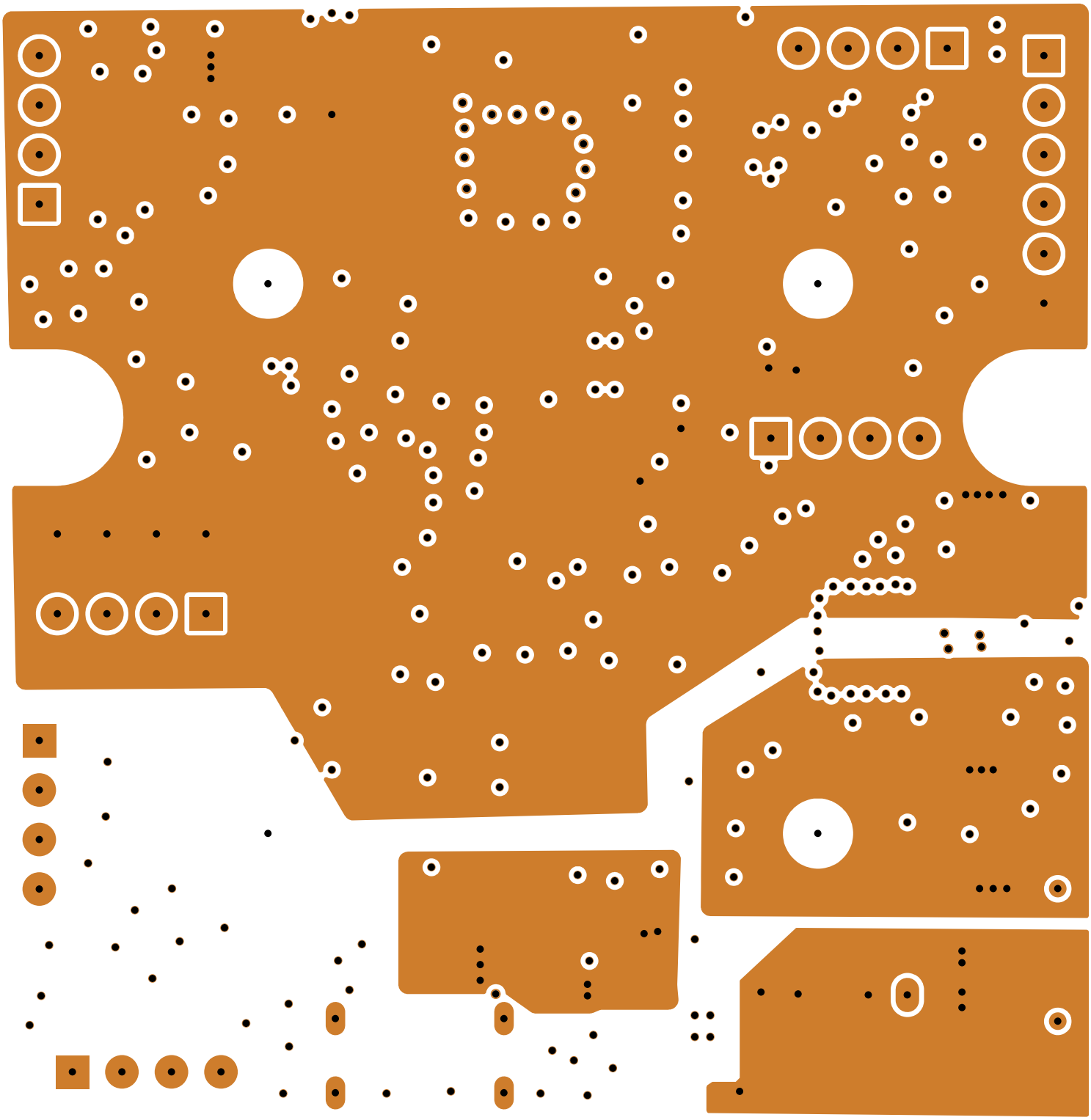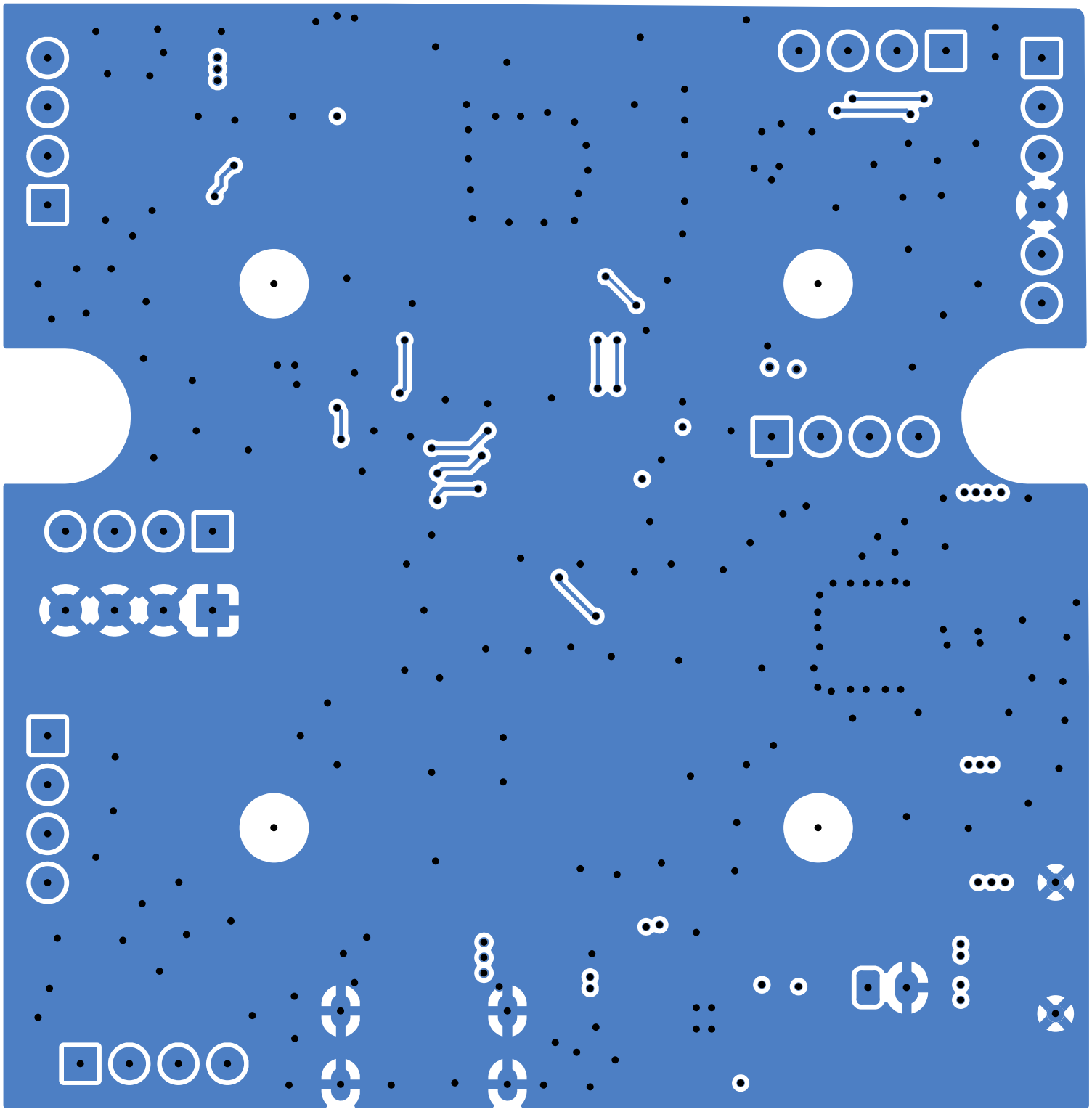
Buck Boost Converter
SW1
MST-12D18G4 SPDT SLIDE SWITCH
+BATT
VCC
U1
TPS63020DSJT
PGND
PG
PS/SYNC
EN
VIN
VIN
L1
VINA
GND
FB
VOUT
VOUT
L2
L2
L1
C1 22uF 25V
C2 10uF 25V
C3 100nF 16V
L1 1.5uH
R1 1MΩ
R2 180kΩ
C6 10uF 25V
C4 22uF 25V
C5 22uF 25V
+3V3
GND
ESP32
U5
ESP32-WROOM-E
C14 22uF 25V
C15 1uF 50V
C16 100nF 16V
R18 10kΩ
EN
EN/CHIP_PU
32K_XP/GPIO32/ADC1_CH4
32K_XN/GPIO33/ADC1_CH5
MTDI/GPIO12/ADC2_CH5
MTCK/GPIO13/ADC2_CH4
MTMS/GPIO14/ADC2_CH6
MTDO/GPIO15/ADC2_CH3
GPIO34/ADC1_CH6
GPIO35/ADC1_CH7
SENSOR_VP/GPIO36/ADC1_CH0
SENSOR_VN/GPIO39/ADC1_CH3
SENSE
3V3
GND
GND_THERMAL
U0TXD/GPIO1
U0RXD/GPIO3
GPIO0/BOOT/ADC2_CH1
ADC2_CH2/GPIO2
ADC2_CH0/GPIO4
GPIO5
GPIO16
GPIO17
GPIO18
GPIO19
GPIO21
GPIO22
GPIO23
DAC_1/ADC2_CH8/GPIO25
DAC_2/ADC2_CH9/GPIO26
ADC2_CH7/GPIO27
ESP32-WROOM
TXD
RXD
IO0
LED
IO4
IO5
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
Battery IC Charger
+5V
U3
BQ24092DGQR
IN
ISET
ISET2
PRE-TERM
NC
OUT
PG
CHG
TS
EP
VSS
C11 22uF 25V
C12 1uF 50V
R7 1kΩ
R8 2kΩ
D1 Red
R10 1.5kΩ
R9 47kΩ
+BATT
CONNECTORS
J2 Conn_01x04_Pin
J3 Conn_01x04_Pin
J8 Conn_01x04_Pin
J6 Conn_01x04_Pin
J4 Conn_01x04_Pin
J5 Conn_01x04_Pin
J9 Conn_01x04_Pin
J1 Conn_01x06_Pin
PWR_FLAG
J7
R16 5.1kΩ
R17 5.1kΩ
RTS
RXD
TXD
DTR
PROGRAMMING
BC847 Q1
BC847 Q2
R4 10kΩ
R5 10kΩ
R6 10kΩ
C7 100nF 16V
USB POWER
USB1
TYPE-C 6P
U4
0603L100SLYR
PWR_FLAG
+5V
GND
EH
A12
A9
B5
A5
B9
B12
R12 5.1kΩ
R11 5.1kΩ
USER LED
D2 Green
R15 1kΩ
ADC BATTERY LEVEL SENSE
VCC
R13 100kΩ
R14 100kΩ
C13 100nF 16V
SENSE
Sheet: /
File: newesp.kicad_sch
Title: Smart Home ESP32 with Battery Power
Size: A4
Date: 2026-03-16
Rev:
KiCad E.D.A. 10.0.1
Id: 1/1